

Zhengkao He

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RESEARCH INTERESTS

- Diffusion language models and efficient LLM inference
- Test-time reasoning, parallel decoding, and inference-time scaling
- Mechanistic interpretability and inference-time control of language models

EDUCATION

University of Virginia

Charlottesville, United States

Doctor of Philosophy in Computer Science (3.9 / 4.0 GPA)

Aug 2024 – May 2029 (expected)

- Advisor: Prof. Aidong Zhang

Tongji University

Shanghai, China

Bachelor of Engineering in Computer Science and Technology (88.50/100 GPA)

Sept 2020 – July 2024

- Relevant Coursework: Machine Learning, Deep Learning, Data Mining, Computer Vision, Algorithms.

PUBLICATIONS

- **Zhengkao He**, Bohan Liu, Guangzhi Xiong, Aidong Zhang. “Reliable Parallel Decoding in Masked Diffusion Language Models.” *arXiv Preprint*, 2026.
- **Zhengkao He**, Guangzhi Xiong, Sanchit Sinha, Bohan Liu, Wenqian Ye, Aidong Zhang. “Rethinking Reasoning Paths as Phase-Structured Trajectories.” *arXiv Preprint*, 2026.
- **Zhengkao He**, Guangzhi Xiong, Boyang Liu, Sanchit Sinha, Aidong Zhang. “CASL: Concept-Aligned Sparse Latents for Interpreting Diffusion Models.” *The Fortieth Annual Conference on Neural Information Processing Systems. (NeurIPS)*, 2026.
- **Zhengkao He**, Guangzhi Xiong, Bohan Liu, Sanchit Sinha, Aidong Zhang. “Triggering Chain-of-Thought via Latent Feature Interventions in Large Language Models.” *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP Main)*, 2026.
- **Zhengkao He**, Sanchit Sinha, Guangzhi Xiong, and Aidong Zhang. “GCAV: A Global Concept Activation Vector Framework for Cross-Layer Consistency in Interpretability.” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 614-623.
- Guangzhi Xiong, **Zhengkao He**, Bohan Liu, Sanchit Sinha, Wenqian Ye, Aidong Zhang. “Capturing In-Context Learning Dynamics with Task Operators” *The Fortieth Annual Conference on Neural Information Processing Systems. (NeurIPS)*, 2026.
- Bohan Liu, Wenqian Ye, Guangzhi Xiong, **Zhengkao He**, Sanchit Sinha, Aidong Zhang. “Towards Robustness against Typographic Attack with Training-free Concept Localization”, *European Conference on Computer Vision (ECCV)*, 2026.
- Guangzhi Xiong, Sanchit Sinha, **Zhengkao He**, Aidong Zhang. “Retrieving Counterfactuals Improves Visual In-Context Learning.” *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- Guangzhi Xiong, **Zhengkao He**, Bohan Liu, Sanchit Sinha, Aidong Zhang. “Toward Faithful Retrieval-Augmented Generation with Sparse Autoencoders.” *International Conference on Learning Representations (ICLR)*, 2026.
- Sanchit Sinha, Guangzhi Xiong, **Zhengkao He**, and Aidong Zhang. “Concept-RuleNet: Grounded Multi-Agent Neurosymbolic Reasoning in Vision Language Models.” *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2026. **Oral Presentation**.
- **Zhengkao He**, Ruifan Chen, Yayue Hou, Fei Xie, Xiaoliang Gong, and Anthony G. Cohn. “A Music Labeling Model Based on Traditional Chinese Music Characteristics for Emotional Regulation.” *ICSCA*, 2023.

- Fei Xie, Xiaoliang Gong, **Zhenghao He**, Tongqi Wu, Yan Lu, and Mohan Zhao. “Alzheimer's Disease Early Diagnosis Based on Resting-State Dynamic Functional Connectivity.” *ICSCA*, 2023.
- Mohan Zhao, Lu Pang, Yan Lu, Fei Xie, **Zhenghao He**, Xiaoliang Gong, and Anthony George Cohn. “Conditional Domain Adaptation Based on Initial Distribution Discrepancy for EEG Emotion Recognition.” *MICCAI workshop*.
- **Zhenghao He**. “Comparison Of Different Machine Learning Methods Applied To Obesity Classification,” *MLISE*, 2022.

WORK EXPERIENCE

Alibaba Cloud
AI Model Optimization Intern

Shanghai, China
Mar 2024 – Aug 2024

- Conducted a technical survey on GPU reliability and failure prediction methods in large-scale distributed clusters, reviewing monitoring approaches and machine learning techniques for anomaly detection.
- Analyzed cluster monitoring and telemetry data, including utilization, temperature, and memory metrics, to understand common operational patterns and potential indicators of GPU failures.
- Investigated AI agent architectures for adaptive task scheduling and multi-model orchestration, summarizing recent research and industry practices in cloud-based AI platforms.

Digital Center, HaaenClean Group
Digital Engineer Intern

Shanghai, China
Sep 2023 – Dec 2023

- Built and deployed an anomaly detection pipeline for multi-sensor industrial data.
- Developed an on-premise LLM-based system for secure document retrieval and automated report generation.

Shanghai Research Institute for Intelligent Autonomous Systems
Research Assistant for Assoc Prof. Zhongpan Zhu, Tongji University

Shanghai, China
Jun 2022 – Sep 2022

- Supported the development of research materials and technical presentations on bionic robotics.
- Proposed a patent “Algorithm of Generating Heat Map for Driver’s Attention”, focusing on computer vision-based driver monitoring systems.

HONORS & AWARDS

3 rd Prize (School-level), 2023 Excellent Student Scholarship at Tongji University	Dec 2023
2 nd Prize (National Level), 2022 Embedded System Design Invitational Contest (Intel Cup)	Aug 2022
Bronze Medal (University-level), China College Students Internet+ Innovation and Entrepreneurship Competition	Aug 2022

OTHERS

Technical Skills: Python, PyTorch, TensorFlow, CUDA, C/C++, MATLAB; Git, Linux, LaTeX
Languages: Chinese (Native), English (Advanced)